Proceedings of ICFTBA 2025 Symposium: Global Trends in Green Financial Innovation and Technology
DOI: 10.54254/2754-1169/2025.BJ29289

Financial Market Trend Forecasting using Text Sentiment Analysis: Social Media, News and Economic Texts with Transformer-based Neural Networks

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Abstract. The emerging hegemony of digital information in the decision-making of financial information has changed the way investors are deciphering the signals in the market. In the world where gigantic quantities of text are produced daily with the help of social media, the news, and economic reports, it is important to understand the emotion that is held within such text and foresee the action of the market. The present research is dedicated to the financial market trend prediction on the basis of text sentiment analysis, the way in which the tone of emotion and opinion of the textual data can be linked to stock market changes. This study uses high-end transformer-based models like FinBERT and FinSentGPT to extract and measure the sentiment of various sources of text. The research questions will be addressed in the study: (1) How do the sentiments of varied text sources impact the trend in financial markets? (2) Is the addition of textual sentiment to quantitative indicators a better way of predicting the market? Due to multi-source data collection, preprocessing, and hybrid predictive modeling, it is shown that sentiment positively supports market prediction. It is concluded that the sentiment of text offers important information about investor psychology, which can be used to make more accurate and explainable financial forecasting models.

Keywords: Financial forecasting, sentiment analysis, FinBERT, transformer models, social media

1. Introduction

Digital information has emerged as a key market sentiment and investment behavior, most importantly in the contemporary financial world. As the online communication platforms expand at a frenzied pace, the importance of the textual information within the social media, financial news, and economic reports has become the key determinant in the investor decision-making and the fluctuation of the market. The research subject is the financial market trend prediction by use of text sentiment analysis with the objective of seeking to know the relationship between the emotional and opinion of textual information and financial market movements. The progress of natural language processing (NLP) in the recent past due to the creation of transformer-based models, including FinBERT and FinSentGPT, has significantly increased the precision of the context-based financial

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sentiment extraction [1]. The study utilizes a multi-source approach of text gathering and sentiment mining, which is followed by predictive modeling with the help of machine learning algorithms. It attempts to respond to two important questions, namely, (1) How do text-based sentiments of different sources affect financial market movements, and (2) Can sentiment indicators be used with quantitative financial data to enhance predictive performance? The research has potential, as it could advance behavioral finance and computational linguistics by providing useful tools to investors, analysts, and policymakers to predict change in the market as a result of the use of data-driven sentiment analytics [2,3].

2. Literature review and theoretical background

2.1. Behavioral finance and market psychology

In this section, the researcher discusses the impact that investor emotions and psychological biases have on making financial decisions. Herding, overconfidence, and prospect theory are examples of behavioral finance theories that describe the application of collective sentiment to push the price of assets (excluding their fundamental value) [4]. Market information, especially fear and greed, will trigger emotional reactions, often leading to overreactions and panic selling and can be identified using sentiment analysis. Such dynamics allow us to have the conceptual background of connecting text sentiment and changes in price. Research that shows that volatility and abnormal returns are associated with bursts of social media optimism or pessimism [5]. Therefore, behavioral finance is the conceptual basis of applying textual information and modeling investor mood and predicting price fluctuations in the short term [1,2].

2.2. Evolution of sentiment analysis in finance

This subsection follows the technological history of sentiment analysis, starting with primitive lexicon-based systems and moving on to sophisticated transformer-based systems. The first type of financial sentiment was the predefined word dictionaries, which did not understand the context and mostly falsely interpreted financial terms such as “downgrade” or “short” [6]. With the advent of machine learning, it became more accurate because it allowed pattern recognition that was data-driven. The recent achievements in the field of natural language processing (NLP), specifically with FinBERT and FinSentGPT, have altered the ratio of financial sentiment analysis, as they capture the language peculiarities of the domain [7]. The models have achieved state-of-the-art performances in terms of the extraction of investor tones and the price direction forecasting [3]. In this part, the author describes the fact that deep learning has made it possible to shift the paradigm of contextualized sentiment modelling in financial forecasting.

2.3. Sources of financial tex data

In the next sub-subsection, primary textual data sources that can be used in financial markets in terms of news reports, social media posts, and economical publications will be divided and evaluated. The feelings of retail investors are articulated on websites like Twitter and StockTwits, whereas formal and objective sentiment is provided by the numerous financial news outlets like Reuters and Bloomberg [5]. Studies have shown that news sentiment has been established to be more predictable and reliable in future prediction and that social media sentiment is a reliable indicator of short-term market responses [8]. Economic announcements, such as central bank announcements, are susceptible to the macro-level market expectations. The integration of these

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heterogeneous sources of data is useful in stabilizing sentiment modeling [2]. In this section, the author explains the effectiveness of multi-source integration in the strength and cohort of sentiment-based prediction models [2].

2.4. Research gaps and conceptual framework

The gaps in existing literature that are identified in this section are the lack of cross-source comparisons, the inability to verify causality, and the difficulty of managing data noise. Most of the past studies were frequently targeting individual data sources and neglecting the fact that news and social media sentiment may be complementary [3]. The other constraint is interpretability, as most models are accurate and lack the ability to explain the influence of sentiments [7]. This paper suggests a theoretical framework that combines the behavioral finance theory with transformer-based sentiment models in order to realize multi-source textual signals. The framework focuses on time alignment, sentiment aggregate and hybrid predictive modeling. It functions on the hypothesis that structured and unstructured sentiment fusion increases accuracy and causal validity in financial forecasting [1,2].

3. Data sources and preprocessing

3.1. Data acquisition

This section outlines the methods of the gathering of textual information in financial news APIs, social media sites, and institutional publications. The sources of information in Reuters, Bloomberg, and Twitter will be acquired through approved APIs, thus providing compliance with the ethics and high sourcing [5]. These data will be supported with the economic reports of the World Bank and IMF to capture the macroeconomic sentiment [2]. Automated crawlers and scrapers will be used to collect the feedback of the real-time updates used to indicate the shifting sentiment trends to the display. The storage, security issues and privacy are also the concern of the section to maintain the integrity of the data. In order to diversify the usage of data, this study will involve different aspects of sentiment that generally affect financial markets [1,6].

3.2. Data cleaning and preparation

This will be through data cleaning and at this stage, the irrelevant symbols, advertisements and duplications will be removed with the aim of adding value to text. The preprocessing steps to facilitate textual representation are tokenization, lemmatization, and stop-word removal [9]. It is crucial that the social media data be minimized in noise because they typically include sarcasm, emojis, or useless conversations [5]. This section also covers the non-English data detection and translation as well as linguistic consistency. Text data of clean and cleaned text can thus be the bricks to building effective and strong sentiment prediction models [7,8].

3.3. Time alignment and data structuring

The time alignment guarantees that the extracted sentiments are in fact accurate with the financial price movements. Textual information is usually prior to or aftermarket responses; therefore, by matching timestamps, causal inference and model accuracy can be achieved [1]. The capture of immediate and delayed effects will be done through the techniques of rolling windows and event-based synchronization [10]. The data will be organized as time-series formats that can be used with

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LSTM and transformer models, which will be consistent in time. Imputation and smoothing will be employed to address the issue of missing data and outliers. This correspondence is essential in an attempt to prevent incorrect correlations and to confirm the real sentiment influence on market trends [1,5].

3.4. Data annotation and labeling

This subsection describes the sentiment labeling process that is required to train supervised models. Initial classification will be done by automated labeling based on pre-trained FinBERT predictions and after that, human validation will be done to ensure accuracy [7]. There will be labeled data that consists of sentiment polarity, which is positive, negative, and neutral and confidence scores. Financial professionals will also look through a sample of labels in order to check domain correctness [3]. Besides, distant supervision methods will increase labelled datasets with the help of financial lexicons inciting keyword heuristics [1]. Annotated data enhance the generalization of models and the decrease that is caused by noise-related misclassifications. This part will provide the transparency of the assignment and validation of sentiment labels used in training downstream models [8,9].

4. Sentiment analysis methodology

4.1. Model selection and architecture

The reasoning behind the selection of FinBERT and FinSentGPT as the key models is explained in this subsection because they have better contextual understanding of financial fields [1]. The FinBERT is adapted to comprehend financial news and reports, whereas FinSentGPT is dynamically adapted to conversational information such as social media posts [7]. It further compares them with the classical algorithms such as Naive Bayes and SVM and explains why the transformer is better than the classical algorithms. It also provides us with the architectural layers with attention mechanisms, token embeddings and classification heads. The combination of these two models leads to adaptability of multi-source, which enhances the accuracy of sentiment mining between different categories of text [3,9].

4.2. Fine-tuning and training procedures

In the subsection, the fine-tuning of the pre-trained models, which is facilitated by domain-specific datasets, is explained. The data is divided into training, validation, and test data to ensure that the model is generalized [6]. Performance is optimized by hyperparameter optimization, like the number of batch sizes, learning rate and sequence length. Transfer learning implies that the model relies on prior knowledge and offers corrections to financial situations [1]. The backpropagation of sentiment-labeled data is used as the basis of training and the accuracy of the context is enhanced. Measures of regularization and dropouts are also mentioned in the section that allows avoiding overfitting. The models are fine-tuned and domain-specific accuracy is achieved when it comes to generating sentiment patterns that can be applied to financial prediction [7].

4.3. Sentiment feature extraction

This section describes the conversion of sentiment to quantitative characteristics that can be used in predictive models. The obtained features are sentiment polarity, sentiment intensity, subjectivity, and

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uncertainty indices [1]. High-frequency sentiment signals are smoothed with such methods of aggregation as moving averages and event windows [1]. Normalization of features makes the features comparable across text sources. The resultant variables reflect the trends of market moods with time. Previous literature demonstrates that aggregated sentiment measures are better predictors of past price changes as they indicate the trend of investor emotion change over time [8]. These organized characteristics are key information for the hybrid sentiment-financial prediction models [2,7].

4.4. Model evaluation

Both statistical and financial measures will be used to measure the model performance. The classification quality will be measured using accuracy, F1-score, and AUC, whereas the practical use will be evaluated using economic metrics such as directional accuracy and Sharpe ratio [1]. The added value of transformer-based sentiment extractions is its comparison with the traditional models based on the same platform [3]. The visualization tools, such as confusion matrices and attention maps, will be used to decipher model predictions [9]. The cross-validation will ensure the cross-dataset generalizability. It is a multi-metric test providing the comprehensive view of the model efficiency with the balance of technical accuracy and financial relevance [2,5].

5. Predictive modeling framework

5.1. Feature engineering

In this section, lagged variables, i.e., the delayed reaction of sentiments, will be constructed and a combination of text-based characteristics and technical indicators will be used [1]. The scaling and the normalization procedures standardize the distributions of features. The strongest predictors can be identified by use of correlation and variance [3]. This step offers a decrease in the redundancy and enhances the knowledge of the model. Research indicates that sentiment-based features are chosen properly and subsequently enhance the accuracy of forecasting considerably since they reveal behavioral clues concealed in textual information [2,9].

5.2. Model development

This subsection is concerned with the creation of hybrid prediction models that mix deep and traditional machine learning. LSTM and Transformer models estimate the temporal dependencies, whereas XGBoost and CatBoost models estimate nonlinear dependencies [3]. A weight of every model will be determined based on cross-validation results. This partially quantitative research design correlates with the recent studies that suggest that text and quantitative information will yield superior predictive outcomes [6]. It is aimed at the creation of a corrective and reflexive real-world market forecasting system [2,7].

5.3. Model validation

The reliability and robustness of models are achieved by model validation using different testing methods. Simulation of real-world time progression will be carried out using rolling-window cross-validation, which will not be susceptible to look-ahead bias [1]. The stability of the models will be tested through out-of-sample testing and noise injection under unforeseeable conditions [5]. Precision, recall, and directional accuracy will be among the metrics that will be used to benchmark

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performance [3]. Sensitivity analyses will also show the most effective sources of sentiment, that is, news or social media. Multilens validation can be used to make predictive power capture authentic economic indications as opposed to spurious correlations [2,9].

6. Results, discussion, and implications

6.1. Quantitative results

In this subsection, the statistical results of model testing are explained. Newer FinBERT-based sentiment models, which are based on transformers, are likely to be more precise and stable than the conventional ML models [3]. Multi-source models are characterized by a better Sharpe ratio and predictability of returns [1]. The directional influence of sentiment on market prices is checked by the Granger causality tests [4]. Similar metrics as AUC and F1-score ensure the robustness of the model on datasets [9]. Such empirical findings validate the predictive ability of sentiment-based models in forecasting the trends of financial markets [2,8].

6.2. Discussion of findings

The final subsection is an explanation of the empirical findings in both theoretical and practical terms. The findings prove the statement that sentiment analysis can be a good method to impress the psychology of investors, which adds to the concepts of behavioral finance [4]. The news sentiment provides strong long-term signals, and social media enhances the opportunity to respond to the occurring events [5]. Gu et al. demonstrate that integrating multi-source data markedly boosts hybrid-model performance, yet lingering weaknesses—misinformation and language drift remain [3]. Overall, ethically and methodologically rigorous sentiment analysis sharpens the accuracy and transparency of financial forecasts [2,7].

7. Conclusion

The conclusion will be a summary of the entire study by drawing a conclusion on how text sentiment derived across several sources, such as news articles, and social media posts, as well as economic reports, can be useful in terms of predicting financial market trends. It will reaffirm that there is a better performance of transformer-based models like FinBERT and FinSentGPT than traditional models at forecasting the financial sentiment because of their contextual interpretation of market language. The section will align these findings to the behavioral finance theories, highlighting that the contribution of investor emotions in the development of prices of assets can be quantified. It will also acknowledge some severe limitations, like misinformation in online information and failure to decipher the model. Finally, the conclusion will propose the topics of future studies, such as the development of a multilingual sentiment model, and the ethics of sentiment-based trading, to ensure that the sentiment analysis remains dependable, transparent, and practical to make accurate financial predictions.

Additionally, this study highlights the radical nature of the current financial analytics using artificial intelligence and natural language processing (NLP). Using FinBERT and FinSentGPT as a part of predictive systems will fill the gap between qualitative, emotion-inspired data and market indicators, creating a comprehensive view of how investors act. These findings highlight the importance of the fact that text sentiment is not a mere reflection of the collective opinion but a quantifiable force behind market volatility and the direction of the trend. Financial institutions, investors and policymakers in turn can use sentiment-based knowledge to improve early warning

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systems, risk management practices and decision support systems. Nevertheless, along with such developments, ethical responsibility arises to promote transparency, fairness, and responsibility in algorithmic predictions to prevent any bias and misinterpret the market or increase misinformation. The next direction in research, in this case, is explainable AI and fairness-conscious sentiment models, which can be interpretable and adapt to changing linguistic and economic conditions.

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